

Quick Commerce SLA Crisis

Finding the Real Cause Behind 85% Late Deliveries

Dark Store Network Redesign — Data Analysis Project

Tools: Python · Pandas · Scikit-learn · Power BI

84.88%	1,000,000	9.57 min	100%
Overall SLA Breach Rate	Total Orders Analyzed	Zepto Avg Delivery Time	Haridwar Breach Rate

1. The Business Problem

Quick commerce companies — Blinkit, Zepto, Swiggy Instamart, Jio Mart and others — have built their entire brand promise on one guarantee: **delivery in 10 minutes**. This is not just a marketing claim; it is the core of their competitive advantage.

When we analyzed **1 million orders** across 12 Indian cities, we found that **848,776 orders** — **84.88%** — **were failing to meet this promise**. Nearly 9 out of every 10 deliveries were arriving late.

The central question of this project: **Why is this happening, and what needs to change?**

2. Our Approach

We tested three possible causes of SLA breaches systematically, using data to rule each one in or out:

Possible Cause	What We Found	Verdict
Dark stores too far from customers	All companies average same distance (~7.75 km)	RULED OUT
Too many orders causing overload	All cities handle same volume (~79,000 orders)	RULED OUT
Operational inefficiency inside stores	Zepto: 9.57 min vs Jio Mart: 22.96 min — same distance	ROOT CAUSE

3. Key Findings

Finding 1: The Hidden Base Delay

Even orders delivered within 0-2 km average **12.76 minutes** — already over the 10-min SLA before distance is even a factor. This reveals a fixed processing delay (picking, packing, dispatch) happening

inside every dark store, regardless of how close the customer is.

Finding 2: Zepto vs Everyone Else

Company	Breach Rate	Avg Delivery Time
Zepto	39.7%	9.57 min
Dunzo	82.4%	14.12 min
Blinkit	87.8%	15.07 min
Swiggy Instamart	91.2%	16.02 min
Big Basket	93.5%	17.98 min
Amazon Now	94.3%	18.94 min
Jio Mart	98.0%	22.96 min

All companies travel the same average distance (~7.75 km). Zepto's advantage is purely operational — faster fulfillment, not closer stores.

Finding 3: City Performance Gap

City	Breach Rate	Avg Delivery Time	Status
Delhi	16.4%	7.14 min	GOOD
Noida	87.9%	15.51 min	HIGH
Mumbai	93.6%	17.51 min	CRITICAL
Jaipur	98.2%	20.55 min	CRITICAL
Haridwar	100.0%	27.54 min	CRITICAL

Delhi outperforms every other city despite identical distances and order volumes. Delhi's infrastructure and operational execution is the benchmark all cities must target.

Finding 4: What Actually Drives SLA Breaches

A Random Forest feature importance analysis quantified the contribution of each factor:

Factor	Importance	Interpretation
City Infrastructure	46.8%	Where you operate matters most
Company Operations	37.0%	How you operate matters nearly as much
Delivery Distance	16.2%	Distance is a minor factor — not the root cause

4. Recommendations

Priority	Recommendation	Expected Impact
URGENT	Audit & fix Jio Mart operations — 98% breach rate is unsustainable	High
URGENT	Add dark stores in Haridwar & Jaipur — 100% and 98% breach cities	High
HIGH	Reduce base processing time — mandate max 3-min pick-pack-dispatch	High
HIGH	Replicate Delhi's operational model in Mumbai & Bengaluru	Medium-High
MEDIUM	Introduce monthly SLA compliance reporting per company per city	Medium

5. Conclusion

The quick commerce SLA crisis is not a geography problem — it is an operations problem. Dark store relocation alone will not solve 85% breach rates. The data clearly shows that companies operating in the same cities, traveling the same distances, achieve dramatically different results based purely on how fast they process and dispatch orders.

Zepto proves it is possible to consistently meet the 10-minute SLA at scale. The path forward is to understand what Zepto does differently, standardize that model across all companies, and simultaneously address the city-level infrastructure gaps — starting with Haridwar, Jaipur, and Mumbai.